239 - Sliding Window Maximum

```
Input: nums = [1,3,-1,-3,5,3,6,7], k = 3
Output: [3,3,5,5,6,7]
Explanation:
Window position                Max
---------------               -----
[1  3  -1] -3  5  3  6  7       3
 1 [3  -1  -3] 5  3  6  7       3
 1  3 [-1  -3  5] 3  6  7       5
 1  3  -1 [-3  5  3] 6  7       5
 1  3  -1  -3 [5  3  6] 7       6
 1  3  -1  -3  5 [3  6  7]      7
```

|res| = |nums| - k + 1

1 3 -1   ~~1~~ 3 -1   queue

3 3 -1   3 3 -1

1 3 1 2 0 5

3, 1   (1, 3, 1)

3, 2   (3, 1, 2)

2, 0   (1, 2, 0)

queue - monotonically not increasing stack

3, 3, 1

3, 2

⓪ Adding element to the queue

1, 3, 1, 2   3, ~~1~~ 2

while q and q[-1] < nums[i]

q.popleft()

① Removing expiring elems not in window

LIFO

q[0] == nums[i - k]:

q.popleft()